Highheads Car Wash System

SYSTEM SOFTWARE DESIGN & PROJECT SPECIFICATION

Document Version: 1.0
Status: Approved Architecture

1. Executive Summary

> The **Highheads Car Wash Management System** is a comprehensive digital platform designed to automate and modernise car wash operations. The platform connects vehicle owners, wash station attendants, and facility managers through a unified real-time workflow. By replacing physical queues with structured online reservations, QR ticket scanning, automated bay assignment, and integrated payment processing, Highheads maximizes station throughput, cuts waiting times, and delivers full operational transparency.

2. Project Objectives

- **Queue Elimination:** Allow customers to pre-book specific time slots and select wash packages remotely.
- **Real-Time Tracking:** Provide live updates on vehicle wash status (`Scheduled` → `In Queue` → `Washing` → `Drying` → `Completed`).
- **Dynamic Bay Routing:** Assign incoming vehicles to available wash lanes automatically upon arrival scanning.
- **Multi-Channel Payments:** Support direct online checkout via Mobile Money, Credit/Debit cards, and cash logging.
- **Business Intelligence:** Equip management with clear metrics on daily revenue, peak hours, and staff productivity.

3. User Roles & System Personas

Role	Primary Responsibilities	Core System Benefits
Customer	Registers vehicles, books wash slots, pays online, and tracks wash progress.	Guaranteed wash times, no line waiting, transparent package pricing, and digital receipts.
Attendant	Scans customer QR tickets at entry gate, updates wash stage status in real-time.	Clear daily bay schedules, simple touch interface, eliminated manual paperwork.
Admin / Manager	Configures wash packages, sets pricing, manages staff shifts, reviews revenue reports.	Complete financial oversight, peak hour load balancing, and operational insights.

4. System Architecture & Modules

Customer Module

- User & Vehicle Profiles
- Interactive Slot Booking
- Package & Add-on Selection
- Payment Gateway Integration
- Live Status Progress Bar

Attendant Station

- QR & Code Ticket Scanner
- One-Tap Stage Controls
- Bay Status Toggle
- Lane Assignment Display
- Completed Wash Log

Admin Dashboard

- Dynamic Package Rates
- Revenue & Sales Analytics
- Staff Shift Management
- Bay & Equipment Monitor
- Customer CRM & Logs

5. Non-Functional Requirements

- **Performance:** Sub-second verification of ticket QR codes and real-time state synchronization across devices.
- **Usability:** Touch-first, high-contrast user interface optimized for handheld devices and outdoor sunlight reading.
- **Security:** Encrypted payment tokenization, Role-Based Access Control (RBAC), and HTTPS secure transport.
- **Scalability:** Modular architecture capable of expanding from a single site to multi-branch operations.

6. Recommended Technology Stack

Layer	Technology Choices	Function
Frontend	React.js / Tailwind CSS	Responsive web application for desktop and mobile devices.
Backend API	Node.js (Express) or Python (FastAPI)	High-performance RESTful API and WebSocket connection engine.
Database	PostgreSQL / Redis	Relational storage for bookings/financials; Redis for queue state caching.
Integrations	Stripe / Mobile Money API / Twilio	Automated payment processing and instant SMS/WhatsApp ticket updates.